

MINSK, Belarus

WMO: 268500

Lat: 53.9287N

Lon: 27.6337E

Elev: 224

StdP: 98.66

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.2	-16.1	-22.4	0.5	-18.7	-19.1	0.7	-15.5	10.0	0.6	8.7	-1.6	1.6	120	0.551

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.9	29.6	20.0	27.9	18.9	26.0	18.1	21.0	27.2	20.0	25.8	19.1	24.6	3.2	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.9	14.1	24.1	17.9	13.2	22.7	17.0	12.4	21.8	61.8	27.6	58.2	25.9	55.1	24.6	24.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.2	7.2	6.3	DB	-23.4	32.0	4.2	1.8	-26.4	33.4	-28.9	34.5	-31.2	35.5	-34.2	36.8	(4)
(5)			WB	-23.7	22.6	4.2	1.0	-26.7	23.4	-29.1	23.9	-31.4	24.5	-34.5	25.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.2	-4.7	-3.8	0.6	7.8	13.3	16.9	19.4	18.2	12.7	6.6	2.0	-2.7	(6)	
	DBStd	9.58	5.77	5.68	4.63	4.41	4.33	3.43	3.11	3.35	3.54	4.20	4.32	5.59	(7)	
	HDD10.0	2016	455	387	293	93	17	1	0	0	15	120	242	393	(8)	
	HDD18.3	4181	713	621	551	318	166	68	24	43	171	363	491	652	(9)	
	CDD10.0	1009	0	0	1	26	118	206	291	254	97	16	1	0	(10)	
	CDD18.3	131	0	0	0	0	9	24	57	39	2	0	0	0	(11)	
	CDH23.3	1122	0	0	0	6	84	187	475	349	21	0	0	0	(12)	
	CDH26.7	273	0	0	0	1	13	32	122	103	2	0	0	0	(13)	
(14) Wind	WSAvg	3.0	3.1	3.1	3.2	3.0	3.0	3.0	2.7	2.6	2.7	2.9	3.1	3.2	(14)	
(15) Precipitation	PrecAvg	677	42	36	41	43	64	80	94	70	57	51	50	48	(15)	
	PrecMax	953	91	82	94	96	136	187	290	255	155	142	105	96	(16)	
	PrecMin	503	8	4	7	2	13	13	10	3	12	2	8	14	(17)	
	PrecStd	98	20	15	18	20	32	41	52	46	31	30	21	18	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.6	7.1	14.9	23.9	28.2	29.8	31.8	32.9	25.9	19.2	12.2	8.1	(19)	
		MCWB	5.8	5.0	8.7	14.2	18.1	20.1	21.1	20.4	16.5	14.3	10.6	7.3	(20)	
	2%	DB	4.1	5.0	11.8	20.4	25.9	27.8	29.9	29.2	23.0	16.1	10.4	6.2	(21)	
		MCWB	3.3	3.7	6.8	12.2	17.3	19.0	20.3	19.8	15.7	12.7	9.0	5.6	(22)	
	5%	DB	3.0	3.8	9.0	18.0	23.2	25.8	28.0	27.1	20.6	14.2	8.9	5.0	(23)	
		MCWB	2.2	2.8	5.2	10.9	15.7	18.1	19.2	18.5	14.9	11.7	7.9	4.4	(24)	
	10%	DB	1.8	2.4	6.8	15.6	21.1	23.8	26.1	24.9	18.8	12.7	7.5	3.3	(25)	
		MCWB	1.2	1.5	4.2	9.2	14.3	17.0	18.4	17.7	14.0	10.8	6.6	2.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.7	5.6	9.5	14.7	19.6	21.6	22.8	22.1	18.1	15.4	11.1	7.5	(27)	
		MCDB	6.3	6.6	13.6	22.3	26.3	27.2	29.4	29.4	23.1	18.2	12.2	7.9	(28)	
	2%	WB	3.4	3.9	7.2	13.0	18.1	20.0	21.3	20.6	16.5	13.6	9.3	5.8	(29)	
		MCDB	3.9	4.5	10.3	19.0	24.2	25.5	27.4	27.1	20.9	15.3	10.2	6.1	(30)	
	5%	WB	2.2	2.7	5.7	11.6	16.8	18.9	20.2	19.6	15.5	12.1	7.8	4.3	(31)	
		MCDB	2.7	3.3	8.1	16.4	22.1	24.1	26.1	25.3	19.3	13.9	8.5	4.8	(32)	
	10%	WB	1.1	1.6	4.3	10.3	15.3	17.7	19.2	18.6	14.6	10.9	6.6	2.7	(33)	
		MCDB	1.5	2.2	6.3	14.2	19.9	22.5	24.7	23.7	17.8	12.6	7.2	3.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	4.3	5.2	7.1	9.9	10.2	9.7	9.9	10.2	9.0	6.6	3.9	3.9	(35)	
		MCDBR	3.7	4.9	10.4	13.9	12.9	12.4	12.6	13.6	12.2	9.0	5.3	3.9	(36)	
	5% WB	MCWBR	3.6	4.1	6.4	7.2	6.2	5.9	5.4	6.0	6.5	6.2	4.3	3.7	(37)	
		MCDWR	3.5	4.5	8.9	11.8	11.7	11.0	11.2	11.2	10.5	8.0	4.7	3.8	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.288	0.322	0.354	0.409	0.400	0.385	0.415	0.416	0.378	0.351	0.330	0.290	(40)	
		taud	2.166	2.122	2.160	2.187	2.285	2.354	2.282	2.275	2.345	2.399	2.282	2.202	(41)	
	Edn at Noon		651	748	811	815	847	864	827	800	786	725	601	569	(42)	
		Edh at Noon	65	95	115	128	123	116	123	117	96	74	60	52	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.68	1.49	2.81	4.09	5.20	5.60	5.24	4.48	3.03	1.52	0.55	0.40	(44)	
	RadStd		0.13	0.28	0.43	0.49	0.53	0.45	0.53	0.41	0.28	0.31	0.08	0.11	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	+0.64	N/A	-190	+151	N/A
(47) Regional (11 neighbors)		+0.48	N/A	N/A	N/A	N/A	+0.75	N/A	N/A	N/A	N/A

Nomenclature: See separate page